

Simulink Model For Real-Time Logging Of Process Data In MATLAB Workspace

Shibendu Mahata

In this project, temperature data is logged in real time in MATLAB workspace by building a model using Arduino IO Library in Simulink. Process parameter measurement carried out for experimental research and analytical testing generally needs to be saved in real time in order to conduct further analysis.

This is particularly relevant in the domain of instrumentation engineering where parameters such as temperature, pressure, flow rate, pH, viscosity, etc, are measured and then saved on the computer in order to analyse the dynamics of the process, or for fault diagnosis. This project is based on Arduino Uno and temperature sensor LM35 IC. The authors' prototype is shown in Fig. 1.

Hardware and construction

The components used in this project are:

Arduino Uno. Arduino Uno is an AVR ATmega328P microcontroller (MCU) based development board with six analogue input pins and 14 digital I/O pins. The MCU features 32kB ISP flash memory, 2kB RAM, and 1kB EEPROM. The board provides the capability of serial communication via UART, SPI, and I2C. It can operate at a clock frequency of 16MHz. In this project, the analogue pin A0 of Arduino is used to read the output voltage (from pin no. 2) of LM35.

LM35. This is a precision IC temperature sensor. Its output voltage is linearly proportional to the temperature (in degree Celsius). LM35 exhibits typical accuracies of $\pm 1/4^{\circ}\text{C}$ at room temperature and $\pm 3/4^{\circ}\text{C}$ over a full temperature range from -55°C to $+150^{\circ}\text{C}$. This sensor provides a sensitivity of $10\text{mV}/^{\circ}\text{C}$. In this project, the Vout pin (pin no. 2) of LM35 IC is connected to analogue input pin A0 of Arduino. Pin numbers 1 and 3 of LM35 are connected to +5V and GND of the

Arduino Uno board, respectively.

The connection diagram of the interfacing temperature sensor with the Arduino board is shown in Fig. 2.

Software

The application program is developed in MATLAB R2014a with the Simulink toolbox. In order to build a Simulink

model, 'Arduino IO Library' is needed, which can be obtained by installing 'Legacy MATLAB and Simulink Support for Arduino package' as explained below:

- Download Legacy MATLAB and Simulink Support for Arduino package from the MathWorks website. Extract ArduinoIO compressed folder
- From this folder, copy the pde folder and paste it in C:\Program Files (or Program files x86)\Arduino\Examples (path may be different depending on the installation directory of the IDE)
- After pasting the pde folder in the correct location, open Arduino IDE. If you have pasted the folder in the correct location, you would find pde in File→Examples→pde
- Open adios code from File→Examples→pde
- Connect the Arduino Uno board to your PC. From the device manager, note the COM port at which the Arduino Uno board is installed
- From Tools menu in Arduino IDE, select the board as Arduino

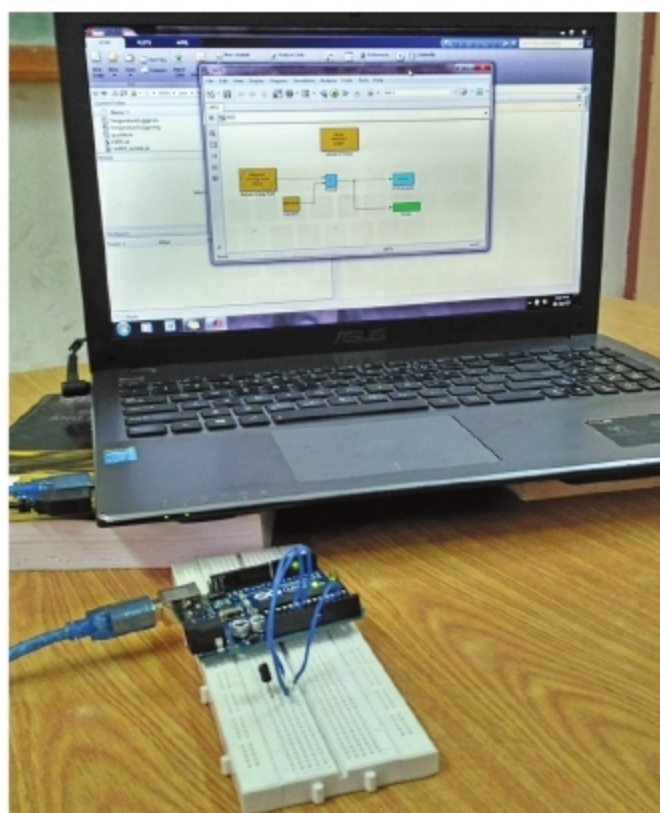


Fig. 1: Authors' prototype wired on breadboard

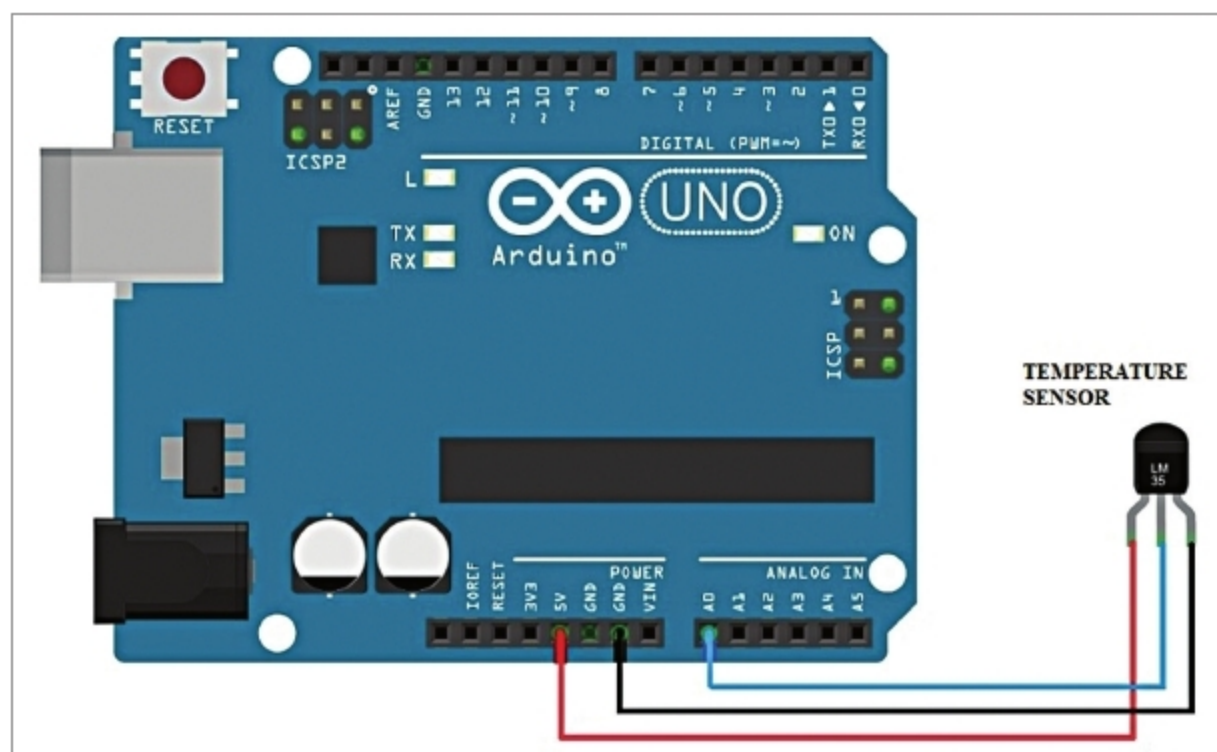


Fig. 2: Connection diagram